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Bumjin Park

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Research

I study the mind of AI through rigorous analysis of neural representations the computational brain of artificial intelligence. My research focuses on Interpretability, Cognitive Architectures, and Neural Representation, exploring how internal structures of AI models reveal the underlying principles of cognition shared between humans and machines.

Ultimately, I aim to develop neural reasoning systems that integrate human and computational forms of reasoning, advancing toward a unified form of general intelligence.

Education

- KAIST (Korea Advanced Institute of Science and Technology) — Sep. 2023 Present
  Ph.D. Student in Artificial Intelligence
  - Advisor: Prof. Jaesik Choi
  - Topic: Integrating Cognitive Architectures into Large Language Models
- KAIST (Korea Advanced Institute of Science and Technology) — Aug. 2023
  M.S. in Artificial Intelligence
  - GPA: 4.17 / 4.3
  - Advisor: Prof. Jaesik Choi
  - Thesis: Partitioned Channel Gradient for Reliable Saliency Map in Image Classification
- Chung-Ang University — Aug. 2020
  B.S. in Mathematics (Double Major in Software Engineering)
  - GPA: 4.39 / 4.5

Publications

- Deontological Keyword Bias: The Impact of Modal Expressions on Normative Judgments of Language Models — ACL Main, 2025
  Bumjin Park, Jinsil Lee, Jaesik Choi, Annual Meeting of the Association for Computational Linguistics, 2025 — Paper | Project
- Memorizing Documents with Guidance in Large Language Models — IJCAI, 2024
  Bumjin Park, Jaesik Choi, International Joint Conference on Artificial Intelligence, 2024 — Paper | Project
- Identifying the Source of Generation for Large Language Models — ICPRAI 2024
  Bumjin Park, Jaesik Choi, Pattern Recognition and Artificial Intelligence. ICPRAI, 2024 — Paper | Project
- Message Action Adapter Framework in Multi-Agent Reinforcement Learning — Applied Sciences, 2025
  Bumjin Park, Jaesik Choi, Applied Sciences, 2024 — Paper

- Cooperative Multi-Robot Task Allocation with Reinforcement Learning — Applied Sciences, 2022
  Bumjin Park, Cheongwoong Kang, Jaesik Choi, Sensors, 2024 — Paper

- Scheduling PID Attitude and Position Control Frequencies for Time-Optimal Quadrotor Waypoint Tracking under Unknown External Disturbances — Sensors, 2021
  Cheongwoong Kang, Bumjin Park, Jaesik Choi — Paper

- Generating Multi-Agent Patrol Areas by Reinforcement Learning — ICCAS / IEEE, 2021
  Bumjin Park, Cheongwoong Kang, Jaesik Choi, International Conference on Control, Automation and Systems (ICCAS) — Paper

Projects

- NYU Global AI Frontier Lab – Interpretability Research — Aug 2024 – Current
  Tools: Python, PyTorch, Transformer Models (Llama, Gemma), Activation Patching, LoRA
  - Mechanistic interpretability of Large Language Models to investigate bias representations.
  - NYU Global AI Frontier Lab is co-directed by Prof. Yann LeCun and Prof. Kyunghyun Cho.

- ADD (Agency for Defense Development) Unmanned Swarm CPS Research Lab — Oct 2021 – Mar 2025
  Tools: ROS, Gazebo, Webots, PyTorch, Python, UAV/UGV Simulation
  - Developed multi-agent reinforcement learning algorithms for patrol and communication tasks in unmanned swarms.
  - Implemented Sim-to-Real transfer using domain adaptation to bridge physical and simulated environments.
  - Built ROS-based communication pipelines between Gazebo and Webots for UAVs (DJI) and UGVs (Husarion Rosbot) [].
  - Achieved 2 journal, 1 conference, 1 domestic journal, and 4 domestic conference publications; filed 2 patents (1 registered).

- Kolmar AlchemyNet & Domain Knowledge (Phase 2) — Nov 2024 – May 2025
  Tools: Python, PyTorch, Transformer, Node.js
  - Advanced Phase 2 of AlchemyNet by incorporating domain knowledge into cosmetic property prediction.
  - Built a web-based AI service for cosmetic composition design and property inference.
  - Submitted 1 journal and 1 conference paper (under review).

- Kolmar AI for Cosmetic Composition (Phase 1) — Aug 2023 – Mar 2024
  Tools: Transformer, Python, Scikit-learn, Pandas, Visualization Libraries
  - Developed AlchemyNet, a neural model inspired by the concept of an alchemist, to encode cosmetic formulations.
  - Predicted multiple physicochemical properties such as viscosity, pH, density, and hardness.
  - Designed explainable embedding spaces linking chemical composition with sensory attributes.

- X-Ray Object Detection and Saliency — Aug 2024
  Tools: Python, PyTorch, Grad-CAM, Explainable AI (XAI)
  - Applied attribution-based XAI methods to improve interpretability of X-ray object detection models.
  - Analyzed decision-making in overlapping object cases to enhance model transparency.

Experiences

- KBS "Insight" (시사기획-창) (Project Leader) — Mar 2026
  Agentic Vulnerability Analysis in AI Agents (Technical Advisor) — [KAIST AI Part | Full Version]
  - Led a 4-person team to analyze real-world vulnerabilities of LLM agents featured in a national broadcast.
  - Conducted indirect prompt injection evaluation, identifying up to 55% attack success rate in GPT-5.2.
  - Developed PnP-XAI App for interpretable LLM behavior analysis and demonstration.
  - Built and analyzed agent systems, OpenClaw and Moltbook, revealing failures in multi-agent environments.

- Visiting Research Scholar – NYU Global AI Frontier Lab — Aug 2024 – Oct 2024
  Mechanistic Interpretability, Bias in LLMs — [ ]
  - Conducted research at New York University's Global AI Frontier Lab (co-directed by Prof. Yann LeCun and Prof. Kyunghyun Cho).
  - Investigated bias representation in large language models through neuron- and circuit-level analysis.
  - Collaborated with researchers on interpretability frameworks for scalable model analysis.

- Teaching Assistant – Time Series, Deep Learning, and Colloquium (KAIST AI) — Sep 2023 – Sep 2025
  Time Series, Deep Learning, and AI Colloquium at KAIST AI. — [ ]
  - Served as a Teaching Assistant for graduate-level courses.

- Lab Representative – SAILAB, KAIST AI — Sep 2023 – Aug 2024
  Focus: Research Organization and Student Coordination — [ ]
  - Served as Ph.D. student representative for the SAILAB research group.
  - Organized internal seminars and managed communication between students and faculty.
  - Coordinated collaboration initiatives and lab activities.

- GPU Server Management – SAILAB, KAIST AI — Jan 2022 – Dec 2024
  Tools: Linux, Docker, Git — [ ]
  - Managed GPU servers and computing resources for SAILAB's research infrastructure.
  - Optimized resource allocation and maintained high-performance computing environments.
  - Automated server monitoring and scheduling with Slurm and Docker containers.